



HOPPER INFORMATION SERIES

SENSOR SUITE

Hopper sensor suite

How an indoor quadrotor measures motion, sees the room, and stays stable without GPS.

NO GPS

INDOOR SENSING

SENSOR FUSION



External sensors see the room; internal sensors track motion

FORWARD RGB CAMERA

- Supplies the Wi-Fi video stream used by Hopper Studio vision.
- It is not part of the Bluetooth flight-control link.

DOWNWARD SENSING MODULE

- Optical flow estimates motion over the floor.
- Time-of-flight range provides a short-range floor-distance cue.

Visible hardware

The exposed underside makes the camera and downward sensor cluster easy to inspect.



INSIDE THE BODY

- IMU: accelerometers + gyroscopes.
- Barometer: air-pressure change.
- Magnetometer: heading cue.
- Temperature sensing and the flight controller support compensation and fusion.

NO GPS / GNSS

No latitude/longitude, global waypoint position, or GPS-derived speed.



What Hopper measures onboard

A IMU

3-axis accelerometer + 3-axis gyroscope: motion, tilt and angular rate.

M Magnetometer

A 3-axis digital compass used by the onboard system for heading cues.

P Pressure altimeter

Air-pressure changes provide a relative altitude trend to the controller.

F Flow sensor

Tracks apparent floor motion to help stabilize movement over textured surfaces.

T Time-of-flight

A short-range downward distance cue to the floor; performance depends on the surface.

C Camera

2 MP articulated camera: 45 deg forward-facing or 90 deg downward-facing.

I Color + IR array

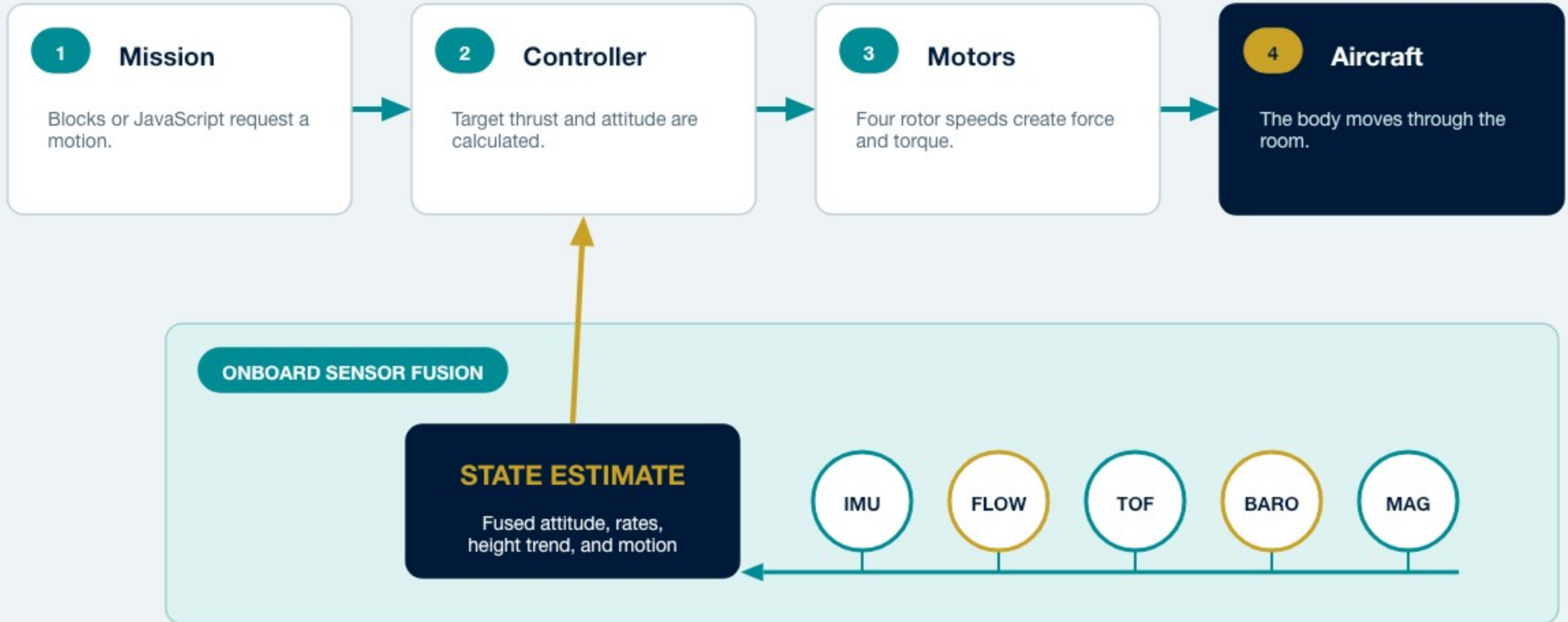
24-bit color and infrared sensing provide optical/environment cues.

D Temperature

Ambient or onboard diagnostic temperature sensing - not a thermal camera.

Onboard sensor does not mean student-readable variable. Hopper Studio exposes only a subset.

Sensor fusion closes the control loop



The controller corrects motor commands from one fused state estimate—not from any single raw sensor.



No GPS is a design constraint

NO GPS

No latitude / longitude

No global waypoint position

Indoor use only

What it can do

- Stabilize attitude with onboard feedback.
- Estimate local motion relative to the floor.
- Use camera targets, objects, paper or AprilTags.

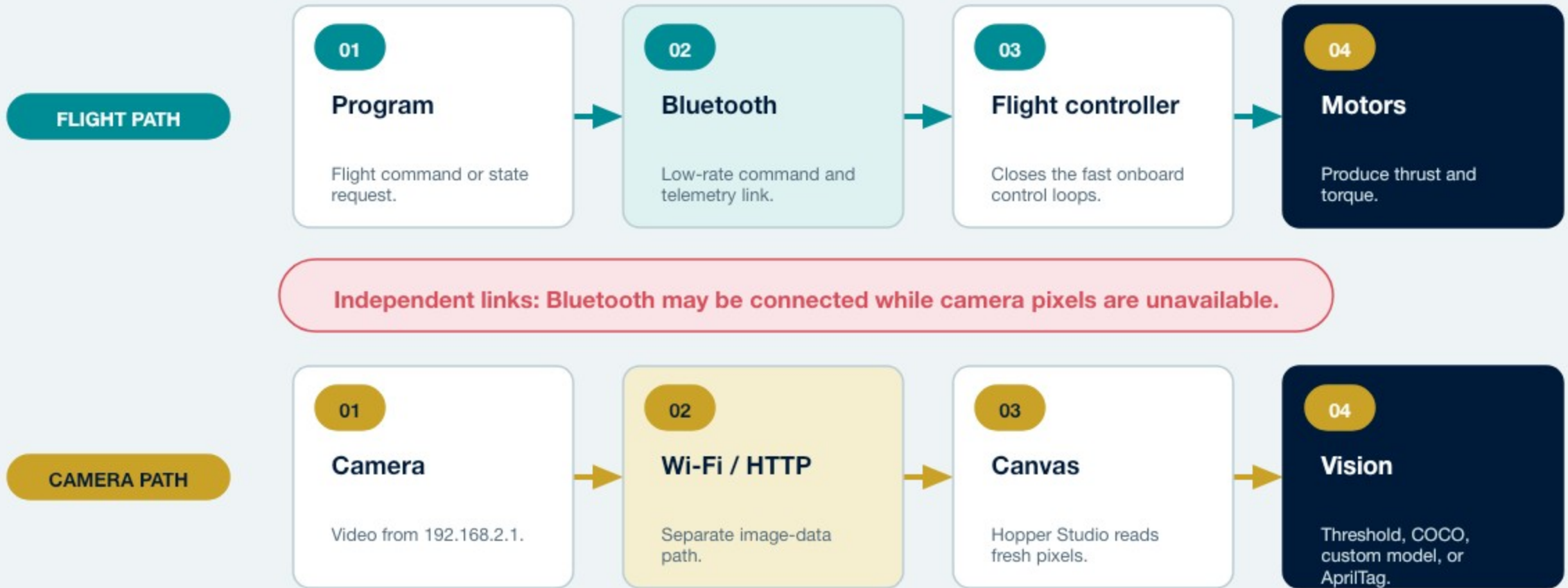
What student code cannot read

- No x/y room position or GPS coordinates.
- No real altitude, velocity or attitude value.
- No raw IMU, range or heading telemetry.

DESIGN MISSIONS AROUND RELATIVE CUES

- Time + power commands: fly forward for 1 second at 15%.
- Visual events: detect a white sheet, object label or tag ID.
- Conservative repeat-and-check logic instead of assumed position.

Camera and communication are separate paths





CLASSROOM CHECKLIST

Preflight for reliable sensing

ROOM + SURFACE

- Fly indoors in a well-lit room.
- Prefer gym or hardwood flooring with visible texture.
- Avoid strong vents, reflective concrete and difficult blue carpet.
- Take off from the floor, not from a hand or table.

APP + MISSION

- Check Bluetooth and Wi-Fi indicators independently.
- Test the exact target under classroom lighting.
- Use repeated observations before a flight decision.
- Keep three feet above obstacles and preserve a landing route.

SOURCES AND FURTHER READING

FTW Robotics - Hopper hardware and ideal flight conditions

<https://ftw-robotics.ai/hopper>

Hopper Studio - local implementation and classroom operating notes

README.md in the Hopper Studio project

Core rule: no GPS, indoor operation, relative cues, conservative decisions.